

NOISE POLLUTION

Noise: Noise is defined as any unwanted, uneasy and undesired sound which disrupts both human and animal comfort. And noise pollution pertains to a situation where there is a lot of noise in the environment.

Noise is measured in Decibel (dB).

Noise Pollution: Noise pollution or sound pollution or environmental noise, is the propagation of noise which has detrimental impacts on the activities of humans and animals. Noise is an irritant and a source of stress. Noise pollution has become a part and parcel of everyday life of urban dwellers. It affects the health physiologically, psychologically and in turn impacting the behavior of the receiver. It is detrimental for the humans, animals, birds and other life forms apart from the plant kingdom. 140 dB is the level when sound becomes unbearable and irritating. This is what compounds to noise pollution.

SOURCES OF NOISE POLLUTION:

1. **Industrial sources:** Rise in noise pollution is the resultant of increased industrialization. Industries like textile mills, engineering plants, printing presses, and metal industries are some examples contributing to noise pollution. Almost all industries incorporating heavy machineries produce very high levels of noise. Equipments such as compressors exhaust fans, grinding mills, and generators increase the overall noise levels in the environment. Workers in these environments are at a huge risk, if not provided with the necessary gears and equipments.
2. **Vehicles for transportation:** In urban setting, the automobile sector is a major contributor to environmental noise. As the population has increased so has the number of vehicles running on road. More often than not people stuck in traffic get impatient and make lot of noise - honking horns, creating nuisance for the neighborhood, commuters and the other people stuck in the traffic just like them. If airports are not situated outside the city airplanes, jet aircrafts taking off and landing can add up to the problem of noise pollution.

3. **Poor urban planning:** Unplanned cities and towns have congested and clustered houses, living spaces and buildings - housing small industries. Thus creating cramped spaces where there is lack of playgrounds, parking spaces, etc. These places are a source of noise from family quarrels, fights between neighbors, wrangles, children playing, and small manufacturing industries, hawkers. Demolition sites and construction sites are other sources where lot of noise is produced due to the use of heavy machines.
4. **Public address gatherings:** Public gatherings as those of rallies, strikes, elections, religious and secular events use addressing systems that are very loud. Such functions normally defy the rules set against public noise pollution. Loud noise from public address systems and music systems during social events such as parties and religious events are another source of noise pollution. The hustle and bustle of open markets produces high levels of noise as street hawkers and shops often use loudspeakers to advertise their merchandise or services.
5. **Agriculture machines:** Heavy machinery and equipment like tractors, harvesters, thrashers, tube wells and drillers are capable of producing sounds as high as 90 dB to 98 dB.

CLASSIFICATION OF SOURCES;

1. **Point Source:** A single identifiable source of noise pollution is known as point source of noise pollution. From a geometric point of view a point source has a negligible extent in comparison to other nonpoint source (or area source). In mathematical modeling considering point sources are so makes the approximation of mathematical point to simplify the analysis. Example: Noise from a jet engine.
2. **Line source:** When a source of air, noise, water pollution or electromagnetic radiation emanates from a linear (one-dimensional) geometry contrasting to a point source, area source, or volume source, is a line source. The most notable linear sources are roadway noise, certain types of noise pollution sources that originate over a range of extent rather than from a distinct point.
3. **Area sources:** Sources of pollution which produce noise from a specified area.

Outdoor and Indoor Noise Pollution:

The Outdoor noise worldwide is largely caused by road traffic, aircrafts, railroads and other vehicles; construction sites and demolition sites; industries etc. Indoor noise pollution can be caused by activities such as plumbing; from machines such as generators, boilers, air conditioners maybe even fans; household equipments like kitchen appliances and vacuum cleaners. Festivities and celebrations happening inside the buildings can sometimes be a source of noise pollution for the neighbors.

NOISE PROPAGATION: Noise is a succession of waves of pressure that travels through compressible media such as air or water (propagation through solid is also possible). For the period of their propagation, waves can be reflected, refracted, or weakened by the medium.

All media have three characteristics which impact the behavior of sound propagation:

1. An association between density and pressure. The speed of sound within the medium is determined by the density – pressure relationship which is affected by temperature.
2. The propagation of sound is dependent on the movement of media say, wind. The medium is in motion the sound is further transported.
3. Medium's viscosity is the deciding factor for the sound weakening rate. For many media like air and water, attenuation because of viscosity is negligible.

In case when sound is travelling through a medium that have varying properties? For example, what happens when sound speed increases with altitude? Sound waves get refracted. They can be focused or dispersed that is how sound levels increase and decrease accordingly.

NOISE MEASUREMENT:

Noise measurement can applied with the intention of gauging environmental noise. Utilized for monitoring of noise at roads, construction sites, airports, industries, recreational venues and communal centers, etc majorly for outdoor noise. The governing laws and limits of environmental noise differ in each country.

Noise is generally measured in a logarithmic scale known as Sound Pressure Level (SPL). Its unit is decibel (dB). Other scales are also used alongside SPL are Sound Power Level (SWL) and Sound Intensity Level (SIL).

SOUND MONITORING: Sound is monitored with equipment called Sound Level Meter.

Distance of Sound Level Meter should be 1.2 meters from the ground. Things that need to be reported while monitoring sound are- background sound level (dB); distance from the source; ambient conditions (temperature, pressure, humidity, wind speed and time of the day). Later on for the diagnosis purpose – sound time history recording also needs to be reported.

EFFECTS ON HEALTH:

Noise is a serious health hazard. Prolonged exposure to noise has many physical, physiological and psychological effects on human health.

Classification of Effects:

- a) **Physical Effects:** If the volume of the noise is high or if the exposure to the noise is repeated, it damages the fine hair cells in the cochlea of the ear, thus causing impairment of hearing (i.e., total deafness). Reduces the sensitivity of ear to pick up small sounds. Normally the hair cells vibrate in response to sound and transmit the message to the brain through auditory nerve. Once the hair cells are damaged permanently by prolonged exposure to loud noise, the person becomes deaf. The body does not replace the damaged hair cells. Besides the sensory cells, the delicate ear drum can be permanently damaged by a sudden loud noise like that of an explosion.
- b) **Physiological and Psychological Effects:** The physiological manifestations of noise pollution are:
 - Headache by dilating blood vessels of the brain, migraine headaches, nausea, dizziness, nervous breakdown, pupil dilation.
 - Increase in the rate of heart-beat, heart palpation.
 - Narrowing of arteries.
 - Fluctuations in the arterial blood pressure by increasing the level of cholesterol in the blood.
 - Digestive spasms due to anxiety.

- Impairment of night vision.
- People may develop signs of aggressive behavior, sleep disturbance, increase in stress level and fatigue.
- Can hamper sleeping pattern.
- Lack of concentration.
- Impairs the mental and creative capabilities which result in reduced work efficiency.

In animals and pets, noise causes them to behave more aggressively than normal, if exposed to loud noise or exposed to loud noise for a longer period of time.

Noise Pollution Level and its Harmful Effects:

Level (in db)	EFFECTS
Up to 23	No disturbance
30—60	Stress, tension, psychological (illness, heart attack) effects especially at upper range.
60—90	Damage to health, psychological and vegetative (disturbance in stomach-gall function, pains in muscles, high blood pressure, disturbance in sleep-ing)
60—120	Damages to health and ontological (ear diseases) effects

Above 120	Painful effects in long run.
-----------	------------------------------

MITIGATION AND CONTROL OF NOISE POLLUTION:

Here are some measures that can be incorporated to reduce noise pollution:

- 1) **Soundproofing:** For the reduction of high frequency sounds from machinery, in Industries and power plants soundproofing is a great option. At homes and buildings that are centers for communal events, installation of soundproofing materials like acoustic foam, sound absorbing foam, acoustic panels, acoustic curtains, moving blankets etc are suggested.
- 2) **Removal of Jarring horns:** Reduction of noise that is related to incessant hoots in traffic, vehicles can move towards installation of horns that have less jarring sounds. In the end it is each people will have to understand their responsibility and keep patience while struck in traffic.
- 3) **Lubrication and better maintenance of machines:** Proper lubrication and maintenance not only improves the efficiency of the machine but it also reduces the sound produced by reducing the friction between the various respective body parts of the machine.
- 4) **Notify Authorities about Disobedience of Noise Rules:** Immediate notification to the concerned authorities if the rules regarding noise control are violated.
- 5) **Stay away from Noisy area:** Industrial areas, power plants, airports, should be located outside the city and town limits. As the noise from these sources can have more harmful implications on the infants, toddlers and senior citizens. Movement of heavy vehicles should be limited inside the city.
- 6) **Follow the Limits of Noise level:** Local response and monitoring bodies should check and take immediate actions against those not following the noise control rules and regulations. Use of loudspeakers for various purposed should be restricted or limited.
- 7) **Control Noise level near sensitive areas:** In ‘Silence Zones’ i.e., areas near hospitals and schools any activity using loud speakers should be restricted. Noise limits boards should be placed near sensitive area.

Go Green by planning trees: Studies suggest that plants and trees can reduce noise by 5 to 10 decibels, they are good sound absorbents. Therefore a green belt surrounding any industrial zone is a good option in mitigating noise pollution.

LEGISLATION RELATING TO NOISE POLLUTION IN INDIA:

The Central Pollution Control Board has formed Committee on Noise Pollution Control.

Guidelines for sources other than industries and automobiles, given by the committee are given as:

- 1) **Public Address System:** License must be taken to use loudspeakers or public address system for any occasion and such address system cannot be used at night between 9pm and 6 am.
- 2) **Bursting Crackers:** Manufacture and sale of crackers causing noise of more than 90 dB at 5 meters distance from the point of bursting are banned. Bursting is prohibited between 10pm and 6am. Bursting is permitted only during festivals.
- 3) **Construction Activities:** Acoustic barriers should be present around construction activities and in industrial areas; the noise level near the construction site will not exceed 75 dB.

NOISE STANDARDS BY CPCB NEW DELHI

Area code	Type of area	Environmental noise standards (L_{eq}) in dBA	
		Day time	Night time
A	Industrial area	75	65
B	Commercial area	65	55
C	Residential area	55	45
D	Silence area	50	40

Notes: Day time shall mean from 6.00 a.m. to 10.00 p.m.; Night time shall mean from 10.00 a.m. to 6.00 a.m.

